

29 Plant Diversity I: How Plants Colonized Land

KEY CONCEPTS

- 29.1** Plants evolved from green algae *p.* 619
- 29.2** Mosses and other nonvascular plants have life cycles dominated by gametophytes *p.* 623
- 29.3** Ferns and other seedless vascular plants were the first plants to grow tall *p.* 629

Study Tip

Make a table: As you read the chapter, make a table of adaptations for life on land that evolved in plants after they diverged from their close aquatic relatives, a group of charophyte algae.

Adaptation	Description	How it facilitates life on land
Cuticle	Waxy covering on outer body surface	Reduces dessication

➔ Go to Mastering Biology

For Students (in eText and Study Area)

- Get Ready for Chapter 29
- Animation: Moss Life Cycle
- Animation: Fern Life Cycle

For Instructors to Assign (in Item Library)

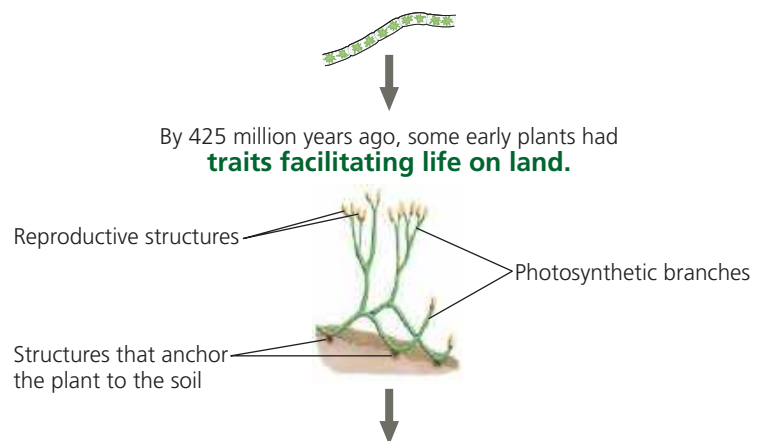
- Visualizing the Concept: Haploid and Diploid Generations Alternate in Plant Life Cycles
- Scientific Skills Exercise: Making Bar Graphs and Interpreting Data



Figure 29.1 For much of Earth's history, the land surface was largely lifeless. Some prokaryotes lived on land 3.2 billion years ago, but it was only within the last 500 million years that small plants, fungi, and animals joined them ashore. Finally, by 385 million years ago, the first forests appeared (but with different species than those of today).

What were the major developments in the evolution of plants?

Plants originated from green algae about 470 million years ago.



Over time, early plants gave rise to a **rich diversity of plants**.

